Skills Summary

Two Ways a Series Argument Goes Wrong

Use this reference while completing your worksheet or when studying.

Skill 1 — When a test reaches no conclusion

Two tests that stay silent, and when. The nth-term test proves divergence when the terms do not approach zero; if they do approach zero it proves *nothing*. The ratio test decides when its limit is not 1; a limit of exactly 1 is no answer at all — and for terms that are a ratio of polynomials it is guaranteed to come out 1, so start with limit comparison instead.

Example: The ratio test returns 1 for $\sum_{n=1}^{\infty} \frac{n}{2n^2+1}$. Settle it another way.

Step 1. Ratio-of-polynomial terms always give a ratio limit of one, so switch to limit comparison and pick the partner from the leading powers, $n/n^2 = 1/n$.

Step 2. $\frac{n/(2n^2+1)}{1/n} = \frac{n^2}{2n^2+1}$, whose limit is finite and positive.

$\Rightarrow$ $L = \mathbf{\frac{1}{2}}$, and since $\sum \frac{1}{n}$ diverges, so does this series

Try it: Find L for $\sum_{n=1}^{\infty} \frac{n}{3n^2+2}$ against $\sum \frac{1}{n}$.

check: $\mathbf{\frac{1}{3}}$, so it diverges

Skill 2 — Endpoints are never free

The ratio test settles the open interval only. At an endpoint its limit equals 1 by construction, so it is silent exactly where you still need an answer. Substitute each endpoint back into the original series and test the ordinary series you get. The two most common results: $\sum \frac{1}{n}$ diverges, so that endpoint is excluded; $\sum \frac{(-1)^n}{n}$ converges, so that one is included.

Example: For $\sum_{n=1}^{\infty} \frac{x^n}{n\,5^n}$ (centre 0, radius 5), test the endpoint $x = -5$.

Step 1. The ratio test cannot decide an endpoint, so substitute the value and identify the ordinary series that results.

Step 2. At $x = -5$ the terms become $\frac{(-5)^n}{n\,5^n} = \frac{(-1)^n}{n}$, the alternating harmonic series, which converges.

$\Rightarrow$ $\mathbf{-\ln 2 \approx -0.693}$, so this endpoint is included

Try it: For $\sum_{n=1}^{\infty} \frac{x^n}{n^2\,4^n}$ (radius 4), test the endpoint $x = 4$ and give the sum.

check: $\mathbf{\frac{\pi^2}{6}}$, so this endpoint is included

Skill 3 — Reading a formula off the wrong term

Formulas read the series as written. A geometric sum is (*first term actually written*) divided by $1 - r$, so a sum starting at $n = 3$ has first term ar^3, not a. An alternating remainder bound is the size of the *first term omitted*, a_{N+1}, not the last one kept. And an integral test decides convergence without giving the sum. Check which index a formula is quoting before you use it.

Example: Find $\displaystyle\sum_{n=3}^{\infty} \left(\frac{1}{2}\right)^n$.

Step 1. The ratio is one half, so the geometric formula applies — but it needs the first term this sum actually contains, which is the one at $n = 3$.

Step 2. First term $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$, ratio $\frac{1}{2}$, so the sum is $\dfrac{1/8}{1 - 1/2}$.

$\Rightarrow \quad \mathbf{\frac{1}{4}}$

Try it: Find $\displaystyle\sum_{n=3}^{\infty} \left(\frac{1}{3}\right)^n$.

check: $\mathbf{\frac{1}{18}}$